

COURSE OUTLINE

Grafana Monitoring & Observability Training



Duration
5 Days · 20 Hours

Format
Live-Online /
In-House ILT

Facilitator
VikashVerma

01. Course Description

This 5-day instructor-led training provides hands-on experience in Grafana Monitoring & Observability. Participants will learn to build dashboards, integrate data sources, configure alerts, and implement Grafana in enterprise environments using industry best practices.

02. Learning Objectives

By the end of this course, participants will be able to —

1. Install, configure, and manage Grafana
2. Create interactive dashboards and visualizations
3. Integrate Prometheus and other data sources
4. Configure alerting, security, automation, and production best practices

DURATION

5 Half Days

LEVEL

Beginner to
Intermediate

LANGUAGE

English

DELIVERY

Online / ILT

04. Prerequisites

- Basic knowledge of Linux/Windows administration
- Familiarity with networking and SQL fundamentals
- Basic understanding of Docker (recommended)

05. Target Audience

- DevOps & Site Reliability Engineers (SREs)
- System, Cloud & Infrastructure Administrators
- Monitoring & Observability Engineers
- IT Operations and Application Support Professionals

06. Module-wise Outline

Day 1 - Grafana Fundamentals & Environment Setup

Module 1: Introduction to Monitoring & Observability

- Why Monitoring Matters
 - Monitoring vs Logging vs Tracing
 - Observability Fundamentals
 - Metrics, Logs and Traces
 - Modern Monitoring Architecture
 - Common Monitoring Tools
 - Grafana Ecosystem Overview
 - Real-world Monitoring Use Cases
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Module 2: Grafana Architecture

- Grafana Components
 - Frontend Architecture
 - Backend Services
 - Data Sources
 - Dashboards
 - Panels
 - Organizations
 - Users & Permissions
 - Plugins
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Module 3: Installation & Initial Configuration

- Installing Grafana
- Docker Installation
- Windows Installation
- Linux Installation
- Configuration Files

- Default Ports
 - Initial Login
 - Folder Structure
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Module 4: Grafana UI Walkthrough

- Home Dashboard
 - Navigation
 - Search
 - Explore
 - Configuration
 - Administration
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Lab 1

- Install Docker
- Install Grafana
- Deploy using Docker Compose
- Login to Grafana
- Explore UI
- Install Plugins

Day 2 - Data Sources & Dashboard Development

Module 5: Working with Data Sources

- What are Data Sources?
- Prometheus
- PostgreSQL
- MySQL
- Elasticsearch
- Loki
- CloudWatch

- **Azure Monitor**
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Module 6: Prometheus Integration

- **Prometheus Architecture**
 - **Metrics Collection**
 - **Targets**
 - **Scraping**
 - **Time Series**
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Module 7: Dashboard Creation

- **Dashboard Design Principles**
 - **Dashboard Layout**
 - **Variables**
 - **Rows**
 - **Panels**
 - **Library Panels**
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Module 8: Visualization Panels

- **Time Series**
 - **Stat**
 - **Gauge**
 - **Pie Chart**
 - **Table**
 - **Bar Gauge**
 - **Heatmap**
 - **Histogram**
 - **Geomap**
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Lab 2

- **Configure Prometheus**
 - **Configure PostgreSQL**
 - **Create Dashboard**
 - **Create Variables**
 - **Add Multiple Panels**
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Day 3 - Querying, Transformations & Alerting

Module 9: Querying Data

- **PromQL Basics**
 - **SQL Queries**
 - **Query Builder**
 - **Explore Mode**
 - **Query Inspector**
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Module 10: Transformations

- **Merge**
 - **Join**
 - **Group By**
 - **Reduce**
 - **Rename**
 - **Organize Fields**
 - **Filter Data**
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Module 11: Alerting

- **Unified Alerting**
- **Alert Rules**
- **Alert Evaluation**
- **Contact Points**
- **Notification Policies**

- **Alert Groups**
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Module 12: Notification Channels

- **Email**
 - **Slack**
 - **Microsoft Teams**
 - **Webhooks**
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Lab 3

- **Create Alert Rules**
 - **Configure Email Alerts**
 - **Configure Slack Alerts**
 - **Test Notifications**
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Day 4 - Security, Administration & Best Practices

Module 13: User Management

- **Organizations**
 - **Teams**
 - **Users**
 - **Roles**
 - **Permissions**
 - **Folder Permissions**
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Module 14: Authentication & Authorization

- **Local Authentication**
- **LDAP**
- **OAuth**
- **Azure AD**
- **Google Authentication**

- **GitHub Authentication**
 - **SSO Concepts**
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Module 15: Dashboard Management

- **Dashboard Versioning**
 - **Import & Export**
 - **JSON Model**
 - **Dashboard Provisioning**
 - **Folder Organization**
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Module 16: Performance Optimization

- **Dashboard Performance**
 - **Query Optimization**
 - **Refresh Intervals**
 - **Caching**
 - **Best Practices**
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Lab 4

- **Create Users**
 - **Configure RBAC**
 - **Import Dashboards**
 - **Export Dashboards**
 - **Optimize Queries**
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Day 5 - Automation, APIs & Capstone Project

Module 17: Grafana HTTP API

- **Authentication**
- **Dashboard API**
- **Folder API**

- User API
 - Data Source API
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Module 18: Dashboard Provisioning

- JSON Provisioning
 - YAML Configuration
 - Dashboard as Code
 - Folder Provisioning
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Module 19: Infrastructure as Code

- Terraform Provider
 - Provision Dashboards
 - Provision Data Sources
 - Automation Best Practices
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Module 20: Production Best Practices

- Backup & Restore
- High Availability
- SSL Configuration
- Reverse Proxy
- Monitoring Grafana
- Troubleshooting